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Data Article

Dataset of personal protective equipment



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ABSTRACT

Personal Protective Equipment (PPE) is critical to the safety and health of workers, especially in high-risk environments such as construction sites and manufacturing facilities. PPE such as helmets and reflective vests play an important role in protecting workers from potential hazards. To protect workers, we have developed a dataset specifically for PPE target detection. The dataset contains 2286 images. These images come from a wide variety of public data platforms and real-life scenarios to ensure a broad representation of applications. In addition, this dataset is annotated according to the official YOLO specification and is seamlessly compatible with popular object detection frameworks such as YOLOv8 and YOLOv11. This dataset improves construction site security and promotes better industrial safety standards.

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Specifications Table

Subject	Computer Science
Specific subject area	Computer Vision for Industrial Safety Personnel Safety Detection: A dataset focusing on the presence or absence of helmets and reflective undershirts can be used as a learning model for industrial safety and computer vision training.
Type of data	Images(.jpg), labels(.txt)

(*continued on next page*)

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Data collection	The dataset was built based on a variety of sources, including public platforms such as GitHub, Kaggle, Roboflow, and real-life photographs taken by authors from different scenarios.
Data source location	GitHub, Kaggle and Roboflow, Country: Taiwan (Hsinchu, Taichung).
Data accessibility	Repository name: Mendeley Data Data identification number: 10.17632/zkzghjvpn2.6 Direct URL to data: https://data.mendeley.com/datasets/zkzghjvpn2/6
Related research article	SRR-YOLOv11: An Enhanced YOLOv11n-Based Model for PPE Detection and Safety Monitoring on Construction Sites, working paper.

1. Value of the Data

- This dataset provides high-quality, well-labelled data focused on the testing of personal protective equipment (PPE) for determining whether PPE is properly worn and in compliance.
- They come from a variety of sources, including public platforms such as GitHub, Kaggle, Roboflow, and photographs taken by authors from different scenes. This collection ensures that the dataset can be adapted to a variety of different scenarios.
- Researchers and developers can reuse this dataset to train models compatible with object detection frameworks such as YOLOv8 and YOLOv11. This dataset covers four categories: Helmet, NoHelmet, NoVest and Vest, which are suitable for detection and identification of PPE.
- The standardised format (YOLO format) allows easy integration of datasets into object detection frameworks such as YOLOv8, YOLOv11, etc., supports training and application of compatible models, and facilitates research extension and innovation.
- The dataset can provide the research community with those working on computer vision, occupational safety, and construction site monitoring.

2. Background

The motivation for creating this dataset arose from challenges identified in existing datasets related to PPE, such as hard hats and reflective vests. Many publicly available datasets were found to have incomplete or inconsistent annotations, making them unsuitable for training reliable machine learning models. In some cases, the annotations were entirely missing, leading to inaccuracies in model development.

To address these issues, this dataset was developed with a focus on completeness and clarity. It includes fully annotated images across four categories: Helmet, NoHelmet, NoVest, and Vest. The data were sourced from platforms like GitHub, Kaggle, Roboflow, and real-world photos taken by authors. This ensures a diverse and representative dataset, designed to support the training and evaluation of computer vision models for industrial safety applications. By prioritizing annotation quality, this dataset aims to overcome the limitations of previous resources and serve as a valuable tool for researchers and developers.

3. Data Description

3.1. Data collection

The dataset was built based on a variety of sources, including public platforms such as GitHub, Kaggle, Roboflow, and real-life photographs taken by authors from different scenarios. The selected photos from the public and photos taken by the authors are listed in Table 1. The criteria to select photos from the public dataset were listed as follows:

(1) The people in the photo must be clearly visible, not blurred or blocked.
(2) The people in the photo must wear helmets or reflective vests.

Table 1
Data collection.

Data source	Number of images
GitHub https://github.com/wujixiu/helmet-detection Apache-2.0: http://www.apache.org/licenses/LICENSE-2.0	1646
Roboflow https://universe.roboflow.com/data-u4eek/helmetvest CC BY 4.0: https://creativecommons.org/licenses/by/4.0/	134
Kaggle https://www.kaggle.com/datasets/niravnaik/safety-helmet-and-reflective-jacket?resource=download Apache-2.0: https://www.apache.org/licenses/LICENSE-2.0	306
Photos taken by authors	200
Total	2286

(3) The people in the photo do not wear helmets or reflective vests.
(4) Photos with one or two people wearing helmets and reflective vests are preferred.
(5) Photos with the background on the construction site or related workplace are preferred.

3.2. Data pre-processing

The dataset removes duplicate and low-quality images to improve the overall quality of the dataset. Follow the standard annotation format of the YOLO model. ALL photos were resized as a resolution of 640×640. Each image is labelled and categorized according to the official YOLO specification, and I have classified the data into four categories: (0) Helmet, (1) NoHelmet, (2) NoVest and (3) Vest. The annotation files are stored in the labels folder and provided in .txt format. The annotation rules are as follows: ⟨class_id⟩ ⟨x_center⟩ ⟨y_center⟩ ⟨width⟩ ⟨height⟩. The ⟨class_id⟩ is the class code; ⟨x_center⟩ and ⟨y_center⟩ represent the normalized coordinates of the center point of the bounding box on the x-axis and y-axis respectively; and ⟨width⟩ and ⟨height⟩ represent the width and height of the bounding box respectively. Fig. 1 is an example for the file 00034_jpg.rf.1c38acedce297348597042dcbfa21a39.txt. The content description is as follows: “0” represents the category of the image as a helmet, “0.646875” is the normalized coordinate of the center of the bounding box on the x-axis, “0.1609375” is the normalized coordinate of the center of the bounding box on the y-axis, “0.19921875” is the width of the bounding box, “0.17734375” is the height of the bounding box The content contains normalized bounding box coordinates and class IDs, which are structured data to facilitate the training and validation of machine learning models in object inspection tasks.

3.3. Training and validation images

The data is divided into two subsets: train and validation, each of which contains an images folder for storing training and validation images. These images cover workers wearing or not wearing PPE in multiple scenarios.

3.4. Training and validation labels

Each subset also contains a labels folder for storing annotation files (.txt format). Each labelled file corresponds to a single image and follows the format specified by YOLO, making it easy to use directly in the YOLOv8, YOLOv11 object inspection frameworks.



Fig. 1. Example of the Data Pre-processing.

3.5. Types of tags

The labels in the dataset were classified into the following four categories: (0) Helmet: workers wearing helmets, (1) NoHelmet: workers not wearing helmets, (2) NoVest: workers not wearing reflective vests, and (3) Vest: workers wearing reflective vests.

The dataset contains a total of 2286 photos with a resolution of 640×640, including 1646 photos from the GDUT-HWD public dataset [1], which was also adopted in Wu et al. [2], 134 photos from the HelmetVest Computer Vision Project public dataset [3], which was also used by Li et al. [4], 306 photos from the Safety Helmet and Reflective Jacket public dataset [5], and 200 real-world photos taken by authors from different scenes.

The total number of images in this dataset is 2286, and does not include augmented images. The images have been randomly assigned as training images (1829 images, 80 %) and validation images (457 images, 20 %) for training and validation of machine learning models. The data shown in Table 2 is not the total number of images, but the number of times each category appears in different images. For example, the Helmet category appears in 1130 training set images, which means that out of the 1829 training set images, 1130 images contain at least one Helmet. Since one image may contain multiple categories (for example, an image contains both Helmet

Table 2
The number of Images.

Category	Training	Validation	Total
(0) Helmet	1130	260	1390
(1) NoHelmet	506	116	622
(2) NoVest	521	140	661
(3) Vest	636	173	809

Table 3
The number of annotations.

Category	Training Annotation	Validation annotation	Total annotation
(0) Helmet	1970	451	2421
(1) NoHelmet	960	214	1174
(2) NoVest	836	222	1058
(3) Vest	1096	289	1385

and Vest), the total number of images in each category will exceed the total number of images in the training set or validation set.

4. Experimental Design, Materials and Methods

The annotation tool used in the study is Roboflow Annotate. The authors manually marked objects in the image and then export them into a YOLO format annotation file (.txt). The process is as follows:

Step 1. Upload all images to Roboflow.
Step 2. Manually mark the bounding box for whether the person is wearing a helmet or a reflective vest, and divide it into four labels: Helmet, NoHelmet, Vest, and NoVest.
Step 3. Double check and make corrections for any errors after the labeling is completed.
Step 4. Automatically generate two folders. The first folder contains the images, and the second folder contains the labeling files (.txt) corresponding to each image through Roboflow system. The file names of these labeling files will be the same as the images, so that they can be correctly matched during model training.
Step 5. Use python to divide the labeled files into train and validation folders, and then divide them into images/ and labels to be directly used for YOLO model training.

Table 3 represents the number of annotations in the four categories. In the training set, there are 1970 annotations in the Helmet category, 960 annotations in the NoHelmet category, 836 annotations in the NoVest category, and 1096 annotations in the Vest category. In the validation set, there are 451 annotations in the Helmet category, 214 annotations in the NoHelmet category, 222 annotations in the NoVest category, and 289 annotations in the Vest category. The number of annotations in the four datasets is not much different, therefore, this is a balanced dataset. The folder contains a configuration file, data.yaml, which is used to specify the location, total number of classes, and name of the training and validation data. The configuration file is structured as follows:

- path: Defines the root directory of the dataset.
- train: Set the path of the training picture.
- val: Set the path of the validation picture.
- nc: Indicates total number of categories (4 categories in this dataset).
- names: Define the name of each category including Helmet,NoHelmet,NoVest and Vest and the corresponding number.

Table 4
The experiment results.

Model	Precision	Recall	mAP50	mAP50–95	FLoPs(G)
YOLOv8	0.593	0.692	0.664	0.474	6.8
YOLOv11	0.589	0.730	0.691	0.483	6.3

The YOLOv8 model is developed based on the YOLOv5 model, which uses anchor-free detection and a new convolutional architecture to improve the accuracy of feature fusion and object detection. YOLO11 uses a new module and deep separable convolution to enhance the ability of feature extraction. It also optimizes GPU computing and parameter simplification to improve detection accuracy and speed. The experiments carried out using this dataset are briefly presented in Table 4. The Precision, Recall, mAP50 and mAP50–95 of the YOLOv8 model are 0.593, 0.692, 0.664 and 0.474, respectively in the detection of the PPE dataset. The Precision, Recall, mAP50 and mAP50–95 of the YOLOv11 model are 0.589, 0.730, 0.691 and 0.483, respectively. The detection levels of the two models are not far apart, but it is worth mentioning that YOLOv11 has an advantage in terms of computational complexity, requiring only 6.3 GFLoPs, which is lower than YOLOv8′s 6.8 GFLoPs.

The structure and annotated format and data within the dataset are directly applicable to the YOLOv8 and YOLOv11 object inspection frameworks, making it a valuable resource for researchers, developers and practitioners. The dataset can be applied to improve industrial safety monitoring systems and enhance construction site safety.

Limitations

This dataset is limited to equipment commonly found on construction sites and does not include protective equipment found in other industrial environments. This dataset covers only four categories: Helmet, NoHelmet, NoVest and Vest. while comprehensive for PPE detection in construction contexts, do not account for other critical safety equipment or scenarios in different industries.

Ethics Statement

The authors have read and follow the ethical requirements for publication in Data in Brief and confirm that the current work does not involve human subjects, animal experiments, or any data collected from social media platforms. The criteria to select photos from the public dataset were under the terms and conditions of the license CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/) and Apache-2.0 (http://www.apache.org/licenses/LICENSE-2.0)

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

The authors declare that generative AI and AI-assisted technologies have not been used in the writing process.

Data Availability

Dataset of Personal Protective Equipment (PPE) (Original data) (Mendeley Data)

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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